

ETL Pipeline for Mining Equipment, Weather, and Production Logs

Overview

This data warehouse integrates data from three main sources to support mining production and weather condition analysis and reporting:

1. **Weather Data** - Collected from an external API and stored in the `raw.weather` table.
2. **Equipment Sensor Data** - Imported from CSV files and stored in the `raw.equipment_sensors` table.
3. **Production Data** - Daily mining production data originating from a MySQL database, loaded into `raw.production_logs` and `raw.mines` tables (mine reference data).

Data from these sources are processed and aggregated into the transform and data mart layers to support aggregated analysis and reporting.

Layers Description

1. Raw Layer

This layer contains raw data extracted directly from the original data sources without transformation or aggregation.

- **raw.weather**: Stores daily weather data including date, temperature, and precipitation.
 - **raw.equipment_sensors**: Stores heavy equipment sensor data including status, fuel consumption, and maintenance alerts.
 - **raw.mines**: Contains master data of mining locations and identities.
 - **production_logs**: Daily production data with details on shift, tons extracted, and quality grade.
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2. Transform Layer

This layer processes raw data to produce cleaned, aggregated, and integrated datasets.

- **transform.equipment_utilization_daily**: Daily summary of equipment utilization based on sensor data.
 - **transform.production_summary_daily**: Daily summary of production combining production logs with mine reference data, including total tons and average quality.
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3. Data Mart Layer

This layer contains tables designed for business analysis and reporting purposes using aggregated and ready-to-query data.

- **data_mart.production_weather_summary_daily**: Daily aggregated table integrating production data with weather data and fuel efficiency for analyzing weather impact on mining production.

Source Data Integration

Source	Type	Raw Table(s)	Import/Extraction Method
Weather API	External API	raw.weather	Data fetched through API calls
Equipment Sensors	CSV File	raw.equipment_sensors	Loaded from CSV into the table
MySQL Database	Relational DB	raw.production_logs , raw.mines	Replicated or ETL loaded into DorisDB

Summary

This data warehouse adopts a layered architecture to ensure well-managed data:

- The **Raw layer** stores the original source data without modifications.
- The **Transform layer** cleanses, calculates, and aggregates data as required for analysis.
- The **Data Mart layer** prepares aggregated, analysis-ready data for dashboards and reports.

Would you like me to provide example SQL queries to move data between layers or generate additional documentation such as a data warehouse architecture diagram?

IP list

Service	Container Port	Host Port	Description
fe	8030	8030	Doris Frontend HTTP API
fe	9030	9030	Doris Frontend HTTP API (alt port)

Service	Container Port	Host Port	Description
fe	9010	9010	Doris Frontend internal communication
be	8040	8040	Doris Backend HTTP API
be	9050	9050	Doris Backend port
mysql	3306	3306	MySQL default port
metabase	3000	3000	Metabase Web UI port

ENV

```

MYSQL_HOST=172.20.0.12
MYSQL_USER=etluser
MYSQL_PASS=etlpass
MYSQL_DB=coal_mining
MYSQL_PORT=3306

```

```

DORIS_HOST=172.20.0.10
DORIS_USER=root
DORIS_PASS=
DORIS_DB=raw
DORIS_PORT=9030

```

Requirements

```

mysql-connector-python
pandas
requests
dotenv

```

Runing

```
docker-compose up --build
```

Dashboard Query

dashboard 1. Daily production trends (total_production_daily)

```

SELECT
    DATE_FORMAT(`date`, '%Y-%m') AS month,
    SUM(tons_extracted) AS total_tons_extracted
FROM
    production_weather_summary_daily
GROUP BY
    month
ORDER BY
    month;

```

2. Quality Grade

```
SELECT
mine_id,
sum(quality_grade*tons_extracted)/sum(tons_extracted) quality_grade
from transform.production_summary_daily
group by 1
[[where {{date_start}} > date and date < {{date_end}}]]
```

3. Scatter Weather Impact

```
SELECT
precipitation_sum,
tons_extracted
from production_weather_summary_daily
[[where {{date_start}} < date and date < {{date_end}}]]
```